

Curriculum Vitae

Employment

2024

Postdoctoral Research Grant Holder, *National Institute of Astrophysics - Astronomical Observatory of Trieste*, Trieste, Italy

2023-24

Postdoctoral Research Fellow, *Tata Institute of Fundamental Research*, Mumbai, India
Department of Theoretical Physics

Education

2019-23

PhD in Physics, *The University of Nottingham*, Nottingham, UK

Supervisors: Dr James S Bolton, Dr Nina Hatch

Thesis: Simulating neutral hydrogen across cosmic time: the 21-cm forest in late reionization models and the global 21-cm signal at cosmic dawn

○ Vice-Chancellor's Scholarship for Research Excellence

○ Graduated July 2023

2014-19

MPhys in Physics, Astrophysics, *The University of Edinburgh*, Edinburgh, UK

○ Graduated July 2019 with the First Class

2016-17

Exchange program, *California Institute of Technology*, Pasadena, USA

○ GPA: 3.3

Observing Proposals

2026

PI of uGMRT Cycle 50 observations (ID: 50.010), *A uGMRT Band-2 Search for the 21-cm Forest 1D Power Spectrum at $z \simeq 5.8$* , 60 hr awarded

Software

2024

21cm-forest_1DPS repository, https://github.com/tomassoltinsky/21cm-forest_1DPS

Forward-modelling the 21-cm forest with instrumental features and compute its 1D power spectrum

Used in 2 publications

Technical and Academic Experience

2025

Refereeing scientific papers, for *A&A*, *PRL*, *PRD*

2023-24

In-Kind Directable Software Development for the LSST-DESC, *Tata Institute of Fundamental Research*, Mumbai, India

○ Development of a code for calculating 3D trispectrum from N-body simulations

2018-19

MPhys Project, *The University of Edinburgh*, Edinburgh, UK

Extreme Variability of Active Galactic Nuclei ([report](#))

Supervisor: Prof Andy Lawrence

○ Tested black hole accretion disc models in the context of extreme variability of AGN luminosity

○ Included a public summary (created a [card game](#))

2018

Cormack Vacation Research Scholarship and Deans' Vacation Scholarship, *The Institute for Astronomy*, Edinburgh, UK

21cm Signal Marking the End of Dark Ages ([report](#), [poster](#))

Supervisor: Dr José Oñorbe

○ Mock observations of global 21cm signal with focus on Dark Ages based on Nyx simulations

2018

Senior Honours Project, *The University of Edinburgh*, Edinburgh, UK

Scalar Field Dark Energy

Supervisor: Prof Andy Taylor

○ Simulated the expansion of the universe driven by dark energy modeled as a dynamical scalar field

2017-18

Telescope Group Project, *The University of Edinburgh*, Edinburgh, UK

Measurement of the Characterising Properties of Exoplanet TrES-2b and its Host Star

Supervisor: Prof Philip Best

○ Transit Photometry and Doppler Spectroscopy methods

○ Observations performed with Edinburgh University's Meade MAX 20in ACF telescope

2017

Summer Undergraduate Research Fellowship, Caltech, Pasadena, USA

Redefining the Habitable Zone

Supervisor: Prof David J. Stevenson

- Determining conditions under which water in liquid phase can be present on the surface of a planet

2017

Research in Experimental Physics, Caltech, LIGO, Pasadena, USA

Signal Processing

Supervisor: Dr Thomas J. Massinger

- Data analysis and investigation of glitches in data

Publications

CD Science Working Group, "Overview of 21cm Experiments at high redshift with SKAO", published in *Advancing Astrophysics with the SKAII (AASKAII)*, 2026 (arXiv:2606.20366). ([link](#))

J Cang, B Ciardi, B Maity, et al., "Exploring the Cosmic Dawn through the 21 cm Forest and High-redshift Radio Sources with the SKA", published in *Advancing Astrophysics with the SKAII (AASKAII)*, 2026 (arXiv:2606.20366). ([link](#))

S K Patil, **T Šoltinský**, S Maitra, et al., "Efficient neutral-IGM inference from noisy 21-cm forest spectra with latent-space U-Net encoding and XGBoost", published in *MNRAS*, Volume 546, Issue 4, id.stag236, 18 pp., 2026. ([link](#))

T Šoltinský, G Kulkarni, S P Tendulkar, et al., "Prospects of a statistical detection of the 21-cm forest and its potential to constrain the thermal state of the neutral IGM during reionization", published in *MNRAS*, Volume 537, Issue 1, pp.364-378, 2025. ([link](#))

E Puchwein, J S Bolton, L C Keating, et al. "The Sherwood-Relics simulations: overview and impact of patchy reionization and pressure smoothing on the intergalactic medium", published in *MNRAS*, Volume 519, Issue 4, pp.6162-6183, 2023. ([link](#))

T Šoltinský, J S Bolton, M Molaro, et al. "Probing quasar lifetimes with proximate 21-centimetre absorption in the diffuse intergalactic medium at redshifts $z \geq 6$ ", published in *MNRAS*, Volume 519, Issue 2, pp.3027-3045, 2023. ([link](#))

T Šoltinský, J S Bolton, N Hatch, et al. "The detectability of strong 21-cm forest absorbers from the diffuse intergalactic medium in late reionization models", published in *MNRAS*, Volume 506, pp. 5818-5835, 2021. ([link](#))

C S Cockell, "Astrobiology: Understanding Life in the Universe", *Wiley-Blackwell*, 2015

- Contributed to and was cited in the Preface to the book

Awards, Scholarship, Grants

2019-22

Vice Chancellor's Scholarship for Research Excellence (EU), *The University of Nottingham*, Nottingham, UK

To support my PhD studies

2019

Students into the World grant, *The Tatra banka Foundation*, Slovakia

2018

RSE Cormack Undergraduate Astronomy Research Prize, *Royal Society of Edinburgh*, Edinburgh, UK

For the "21cm Marking the End of Dark Ages" project

2016

Year 2 Physics Class medal, *The University of Edinburgh*, Edinburgh, UK

For excellent performance during the 2015/16 session

2014

The memorial letter of St. Gorazd, *The Ministry of Education, Science, Research and Sport of the Slovak Republic*, Bratislava, Slovakia

For successful representation of the Slovak Republic at Austrian Young Physicists' Tournament and achievement of bronze medal

2014

Best in Natural Sciences school award, *Pavol Horov Gymnasium*, Michalovce, Slovakia

For representation of the school at various Physics and Maths competitions

Mentoring and Teaching

2021

Mentoring at GROWNi.sk, Slovakia

- Advising students in their career and study decisions

2016

Outreach talks, Pavol Horov Gymnasium, Michalovce, Slovakia

- "From LA through half the globe to simulated universe", Topic: career path and simulations in astrophysics
- "From LA to LA", Topic: gained experience and how to study abroad
- "Science has no borders", Topic: my research experience and ongoing research in the world

2023

Advanced 21-cm Cosmology School and Workshop, National Institute of Science Education and Research, Bhubaneswar, India, Lectures ([link](#))

Service to Academic Community

2023-24

State of the Universe Seminar Series Organizer, TIFR, Mumbai, India

2022

Astronomy Postgraduate Student Representative, The University of Nottingham, Nottingham, UK

- Main point of communication between PG students and staff

2021-22

Lunch Talks Organizer, The University of Nottingham, Nottingham, UK

2019-22

Physics Coursework Marker, The University of Nottingham, Nottingham, UK

2017-21

Student Ambassador, The University of Edinburgh and the University of Nottingham, Edinburgh and Nottingham, UK

- Resided over the information desk and gave tours of the campus and the School of Physics and Astronomy at the Open Days, Post-offer Visit Days, Visiting Students Welcome Event

2018

UUKi Parliamentary Reception for the Go International: Stand Out Campaign, Westminster, London, UK

- Ambassador for The University of Edinburgh

Presentations at International Conferences and Meetings

2026

EAS 2026, Swisstech convention centre, Lausanne, Switzerland, Talks and posters at 3 sessions ([link 1](#), [link 2](#))

2026

Cosmology Seminar, The Institut d'Astrophysique Spatiale, Orsay-Paris, France, Seminar

2026

Cosmology Journal Club, University of Padova, Padova, Italy, Seminar

2025

The Fifth National Workshop on the SKA Project - From precursors to SKAO: shaping the future of Italian radio astronomy, CNR, Bologna, Italy, Talk

2025

Kaba Kada: Exploring the first billion years of the Universe, Oaks Port Douglas Resort, Port Douglas, Queensland, Invited talk and part of panel

2025

ICRAR-Curtin Colloquium, Curtin University, Perth, Western Australia, Colloquium

2025

NAM 2025, Durham University, Durham, UK, Talk and a poster at 2 sessions

2025

The galaxy-IGM connection in the first billion years, IFPU, Trieste, Italy, Talk

2025

Radio Cosmology and Continuum Observations in the SKA Era: A Synergic View, International Centre for Theoretical Sciences, Bengaluru, India, Talk ([link](#))

2024

SKA Cosmology SWG Meeting, Observatoire de la Cote d'Azur, Nice, France, Talk

2024

IFPU colloquium, IFPU, Trieste, Italy, Colloquium

2024

The First Gigayear(s), Hilo Hawaiian Hotel, Hilo, Hawai'i, Talk

2024

The Origin and Evolution of Supermassive Black Holes, Sexten Center for Astrophysics, Sexten, Italy, Talk ([link 1](#), [link 2](#))

2024

Cosmic Dawn at High Latitudes Conference, Royal Swedish Academy of Sciences, Stockholm, Sweden, Talk ([link](#))

2024

State of the Universe, TIFR, Mumbai, India, Seminar ([link](#))

2024

Cosmology in the Alps Conference, Les Diablerets, Switzerland, Talk ([link](#))

2024

Monday Afternoon Talks, MIT Kavli Institute, Cambridge, US, Talk

2024	Pune-Mumbai Cosmology and Astro-Particle meeting , <i>The Inter-University Centre for Astronomy and Astrophysics</i> , Pune, India, Talk and Chaired a session (link)
2024	Astronomical Society of India meeting , <i>IISc</i> , Bengaluru, India, Talk
2024	HEA group meeting , <i>Masaryk University</i> , Brno, Czechia, Seminar
2023	Astrophysics Seminar , <i>University of Helsinki</i> , Helsinki, Finland, Seminar
2023	Galaxy coffee , <i>Max Planck Institute for Astronomy</i> , Heidelberg, Germany, Talk
2023	21-cm Series Seminar , <i>University of Cambridge</i> , Cambridge, UK, Seminar
2023	Colloquium , <i>National Centre for Radio Astrophysics</i> , Pune, India, Colloquium
2022	Lunch talk , <i>The University of Nottingham</i> , Nottingham, UK, Talk
2022	21-cm Series Seminar , <i>University of Cambridge</i> , Cambridge, UK, Seminar
2021	State of the Universe , <i>TIFR</i> , Mumbai, India, Seminar (link)
2021	NAM 2021 , <i>The University of Bath</i> , Bath, UK, Talk
2021	SAZERAC 2.0 , Recorded talk (link)
2021	Lunch talk , <i>The University of Nottingham</i> , Nottingham, UK, Talk
2020	Lunch talk , <i>The University of Nottingham</i> , Nottingham, UK, Talk

Membership in collaborations

2024	SKA Cosmology and Epoch of Reionization SWGs , link Cosmology , link EoR
2024	21-cm Cosmology Group (India) , link
2023	LSST-DESC, Large scale structure WG , provisional membership, link
2019	Sherwood-Relics simulation suite , link

Training/Skills

2023	Foundation HPC-Skills Course , <i>DiRAC</i>
2023	DP0 Virtual Summer School 2023 , <i>Vera C. Rubin Observatory/LSST</i>
Programming	Python, Jupyter notebook, C, $\text{\LaTeX}$, Mathematica, IDL, Radiative Transfer, 21cmFAST, IRAF, SExtractor, Large data analysis, MCMC, OpenMP parallelisation, Slurm